Three names for one condition

Matched size, matching proportions, straight corner-diagonals — and why the sign is a separate matter.

THE CROSS-RATIO · A WORKING NOTE · JUNE 2026

Cut the segment **PQ** at two points C and D, and write the four directed lengths a = PC, b = CQ, c = PD, d = DQ. The rectangle of width $c+d$ and height $a+b$ falls into four pieces, $(a+b)(c+d) = ac + ad + bc + bd$. Two of them — the **complements** *ad* and *bc* — sit off the main diagonal; the other two — the **diagonal-rectangles** *ac* and *bd* — are the ones the diagonal runs through. The cross-ratio is $\lambda = (a/b)\div(c/d) = ad/bc$, and the harmonic value is $\lambda = -1$.

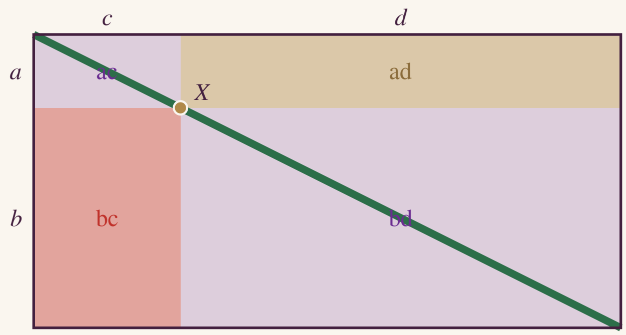

*Figure 1. The dissection at the harmonic cut a = +0.25, b = +0.75, c = −0.50, d = +1.50. Complements ad (gold) and bc (crimson) are the **size** pair; diagonal-rectangles ac, bd (purple) are the **shape** pair. Here |ad|=|bc|=0.38 and ac, bd are both 2:1; the inner point X sits on the diagonal.*

1 · One condition, three names

In its *magnitude*, the harmonic configuration is governed by a single equation: **|ad| = |bc|**. That equation appears in three costumes, and the whole point of this note is that they are not three facts but one.

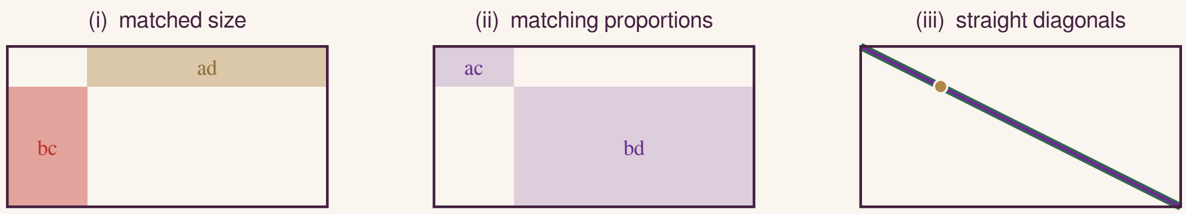

Figure 2. The same harmonic cut, three times; only the emphasis differs.

Matched size. The two complements have equal area: $|ad| = |bc|$ outright. (Euclid I.43 is the geometric form — the complements of the parallelograms about the diameter are equal.)

Matching proportions. The two diagonal-rectangles have equal aspect ratio in the same orientation, $|c|/|a| = |d|/|b|$ (corresponding sides parallel — a rotated copy is excluded), which cross-multiplies to $|ad| = |bc|$.

Straight corner-diagonals. The diagonal of *ac* (top corner to X) and the diagonal of *bd* (X to far corner) are collinear exactly when their direction vectors $(|c|,|a|)$ and $(|d|,|b|)$ are parallel — that is, when $|c||b| = |a||d|$, i.e. $|ad| = |bc|$.

Each statement unwinds to the identical equation. So they are one proposition with three pronunciations: prove any one and the other two are already proved — there is nothing left to check. This is stronger than "any one implies the others." They are the same claim said in the vocabulary of area, of proportion, and of geometry.

Off the harmonic cut, that single equation simply fails — and because it is one equation, all three faces fail with it, together:

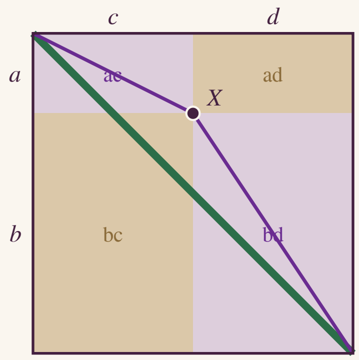

Figure 3. An off-diagonal cut (here λ = 1/3). The complements ad, bc now differ in size; the diagonal-rectangles ac, bd differ in shape; and the two corner-diagonals bend at X, peeling off the green main diagonal. One equation broke, so all three broke at once.

It is worth contrasting this with genuinely independent conditions. "Equilateral" needs two separate equalities, $\text{side}_1 = \text{side}_2$ *and* $\text{side}_2 = \text{side}_3$; knowing only the first leaves you with an isosceles triangle, so you really must check more than one. The trio here is the opposite situation — it is like calling a number "even," "divisible by 2," and "not odd": three phrasings, zero added content. Checking one is not a shortcut past the others, because there are no others.

2 · The second axis: sign

The phrase "harmonic adds one condition on top" lives on a *different* axis, and this is the subtlety. The equation $|ad| = |bc|$ fixes magnitude, which gives $|\lambda| = 1$ — but $|\lambda| = 1$ has two roots: $\lambda = -1$ (harmonic) and $\lambda = +1$ (coincident). The trio cannot tell them apart, because area, proportion, and diagonal-direction are all magnitude quantities. What separates the two roots is the **sign** of *ad* against *bc* — the colour: one cut inside (+), one outside (−).

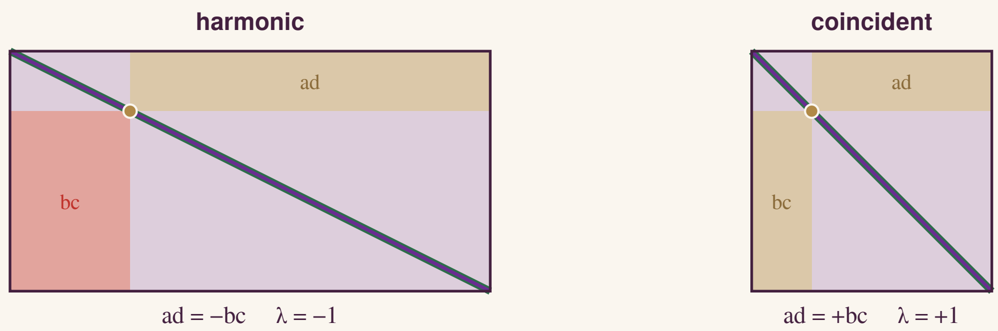

Figure 4. Both cuts sit on the diagonal — matched magnitude, so size, shape, and diagonals all agree, and both panels are drawn to one common scale. They differ only in sign: opposite colours give the harmonic −1; equal colours give the coincident +1 (and C, D collapse to one point, so the box is a square). The trio cannot see this distinction; the colour can.

So the full statement factors into two genuinely independent pieces:

harmonic = (matched magnitude) and (opposite sign) = |ad| = |bc| and ad = −bc.

The first piece is the trio — one condition, three names. The second is the sign flip, which is independent of it (coincident also has matched magnitude, but the *same* sign). "On top" means the sign condition added to the magnitude condition — not a fourth member added to the trio. And it is exactly why the diagrams need *both* the equal-area geometry *and* the gold/crimson colouring to pin down −1 rather than ±1.

3 · Coda: the square root of the minus sign

Everything above renders the minus sign as a colour flip or an arrow reversal. Where does the square root of −1 enter? Push the harmonic relation to its most symmetric form and $\lambda = -1$ becomes a mean-proportional condition, $b^2 = -ac$: the middle length b is the geometric mean of a and $-c$. In magnitudes that is the ordinary geometric mean, $|b| = \sqrt{(|a||c|)}$; but the harmonic signature — one cut inside (+), one outside (−) — forces $b^2 < 0$. The real line holds no number between two oppositely-signed lengths. With $|a| = |c| = 1$ it asks for $b^2 = -1$, and the answer is $b = i$. So i is the geometric mean of +1 and −1 — the mean proportional $1 : i :: i : -1$, which is only $i^2 = -1$ read as a proportion.

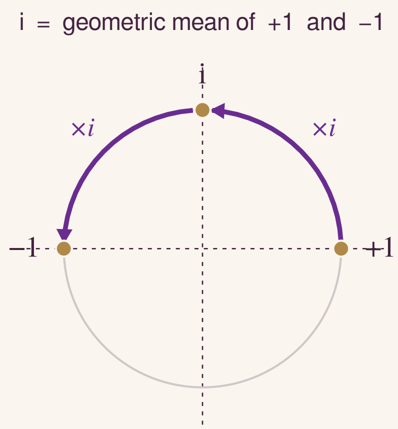

Figure 5. i as the mean proportional between +1 and −1. Each quarter-turn is multiplication by i; two of them make the half-turn, $(\times i)^2 = \times(-1)$.

And the operation behind it: put the frame P, Q at 0 and ∞, and harmonic conjugation becomes $z \mapsto -z$ — multiplication by −1, a half-turn. Its square root is $z \mapsto iz$ — multiplication by i, a quarter-turn — and $(\times i)^2 = \times(-1)$. So i is the square root of the harmonic flip: the quarter-turn whose double is the conjugation, the first step that lifts a point off the real line. The harmonic value −1 itself stays real — which is exactly why honest areas can draw it. The cross-ratio goes properly complex only one step over, at the equianharmonic $e^{\pm i\pi/3}$.

Ledger

Theorem. |ad| = |bc| ≡ the two complements have equal area (Euclid I.43).

Theorem. |ad| = |bc| ≡ the diagonal-rectangles *ac*, *bd* are directly similar — equal aspect ratio, corresponding sides parallel (a rotated copy excluded).

Theorem. |ad| = |bc| ≡ the two corner-diagonals are collinear (parallel direction vectors).

Theorem. The three are mutually equivalent — one proposition, three names, not three independent conditions.

Theorem. harmonic ≡ |ad| = |bc| and ad = −bc; coincident ≡ |ad| = |bc| and ad = +bc. Magnitude and sign are independent axes.

Observation. "Size / shape / diagonals" is a packaging of the single magnitude condition; "matched magnitude + sign" is the full harmonic condition.

Checked. The four statements are equivalent *by algebra* (each reduces to $|ad| = |bc|$); a 3×10^5-cut sweep corroborates this and the conservation law $ad\cdot bc = ac\cdot bd = abcd$, and confirms ad/bc is invariant under 2×10^4 Möbius maps (a cross-ratio) while ac/bd is not.

Theorem. i is the mean proportional between +1 and −1 — $i^2 = -1$ read as $1 : i :: i : -1$, the geometric mean $\sqrt{(1\cdot(-1))}$.

Theorem. With frame $\{0, \infty\}$, harmonic conjugation is $z \mapsto -z$ (half-turn); $z \mapsto i\,z$ (quarter-turn) is its square root, $(\times i)^2 = \times(-1)$. The harmonic value −1 is real; the cross-ratio first goes complex at the equianharmonic $e^{\pm i\pi/3}$.